

Name: Husnain Zafar

Sap id : 70174327

Subject : database system

Group member:

Salman mansha 70174313

Aheed Murtaza 70154171

University Badminton Database

Documentation

Introduction

The University Badminton DBMS (Database Management System) is designed to manage badminton-related activities within a university. The system stores and manages information about players, teams, tournaments, matches, coaches, and scores efficiently.

Objectives

Maintain player records

Manage tournaments and matches

Store match results and scores

Manage team and coach information

Reduce manual record keeping

Improve data accuracy and accessibility

System Overview

The University Badminton DBMS is a centralized database system used by sports departments and administrators to organize badminton activities.

Main Users

Sports Administrator
Coaches
Players
Tournament Organizers

3. Functional Modules

3.1 Player Management Module

Features:

Player registration
Player profile management
Department and semester records
Contact information storage

Functions

Add player
Update player details
Delete player records
Search player information

3.2 Team Management Module

Features

Team creation
Team member assignment
Team captain selection

Functions

Create teams
Assign players to teams
View team details

3.3 Coach Management Module

Features

Coach registration
Coach profile management
Team assignment

Functions

Add coach
Assign coach to team
Manage coach information

3.4 Tournament Management Module

Features

Tournament scheduling
Tournament categories
Match scheduling
Tournament status management

Functions

Create tournaments
Schedule matches
Generate fixtures

3.5 Match Management Module

Features

Match records
Match date and venue
Player/team participation

Match result storage

Functions

Schedule matches

Record match results

Update scores

3.6 Score Management Module

Features

Score entry

Winner declaration

Match statistics

Functions

Enter scores

Calculate winners

Display score history

4. Database Design

4.1 Players Table

Field Name	Data Type	Description
Player_ID	INT (PK)	Unique player ID
Name	VARCHAR	Player name
Department	VARCHAR	Player department
Semester	INT	Current semester
Contact	VARCHAR	Phone number

4.2 Teams Table

Field Name	Data Type	Description
Team_ID	INT (PK)	Unique team ID
Team_Name	VARCHAR	Team name
Captain_ID	INT (FK)	Team captain

4.3 Coaches Table

Field Name	Data Type	Description
Coach_ID	INT (PK)	Unique coach ID
Coach_Name	VARCHAR	Coach name
Contact	VARCHAR	Coach contact

4.4 Tournaments Table

Field Name	Data Type	Description
Tournament_ID	INT (PK)	Unique tournament ID
Tournament_Name	VARCHAR	Tournament title
Start_Date	DATE	Tournament start date
End_Date	DATE	Tournament end date

4.5 Matches Table

Field Name	Data Type	Description
Match_ID	INT (PK)	Unique match ID
Tournament_ID	INT (FK)	References Tournaments table
Team1_ID	INT (FK)	First team
Team2_ID	INT (FK)	Second team

Field Name	Data Type	Description
Match_Date	DATE	Match date
Venue	VARCHAR	Match venue
Winner_Team	INT	Winning team

4.6 Scores Table

Field Name	Data Type	Description
Score_ID	INT (PK)	Unique score ID
Match_ID	INT (FK)	References Matches table
Team1_Score	INT	Team 1 score
Team2_Score	INT	Team 2 score

5. Database Relationships

Players ↔ Teams

Multiple players can belong to one team.

Relationship: Many-to-One

Players (M) ----- (1) Teams

Teams ↔ Coaches

One coach can manage multiple teams.

Relationship: One-to-Many

Coaches (1) ----- (M) Teams

Tournaments ↔ Matches

One tournament can contain many matches.

Relationship: One-to-Many

Tournaments (1) ----- (M) Matches

Matches ↔ Scores

One match has one score record.

Relationship: One-to-One

Matches (1) ----- (1) Scores

6. User Roles and Permissions

Role	Permissions
Admin	Full system access
Coach	Manage teams and players
Organizer	Schedule tournaments and matches
Player	View schedules and results

7. Security Features

Secure login system

Password encryption

Role-based access control

Backup and recovery system

8. Advantages of the System

Centralized badminton records

Easy tournament management

Quick score updates

Reduced paperwork

Improved data management

9. Future Enhancements

Mobile application integration

Online match streaming

Live score updates

AI-based player performance analysis

10. Conclusion

The University Badminton DBMS provides an efficient solution for managing badminton activities within a university. It simplifies tournament organization, player management, and score tracking while ensuring accurate and secure data management.